SMARTPARK ECOSYSTEM

PROJECT BUDGET PLAN REPORT

Project Name	SmartPark – Intelligent Autonomous Parking Ecosystem	**Document Ref.**	SP-BPR-2026-V1.0
Prepared By	Project Engineering Team [Placeholders]	**Target Facility**	500-Bay Commercial Mall Layout
Course / Dept	Mechatronics & Systems Engineering Department	**Submission Date**	November 2026

EXECUTIVE SUMMARY

This report establishes the baseline commercial budget plan for **SmartPark**, an automated autonomous robotic parking facility. The system integrates 4 heavy-duty AGVs, 500 bay occupancy sensors, 6 drop-off/pick-up stations, central fleet management software, a mobile app, and AI predictive modules, alongside value-added EV charging and automated car washing services. Total baseline Capital Expenditure (CAPEX) is **15,000,000 EGP**. Including a 15% Risk Contingency Reserve (2,250,000 EGP), the official Cost Baseline is **17,250,000 EGP**. Annual Operational Expenditure (OPEX) is projected at **2,664,000 EGP**.

1. INTRODUCTION & PROJECT SCOPE

SmartPark eliminates manual vehicle hunting in commercial malls by receiving cars at transfer stations and deploying autonomous vehicles (AGVs) to store cars tightly in high-density bays.

In-Scope Components

- 4 Custom/OEM Heavy-Duty Autonomous Valet AGVs.
- 500 Bay Occupancy & Obstacle Detection Sensors.
- 6 Drop-Off / Pick-Up Transfer Stations.
- Central Fleet Software, AI Engine, & Mobile App.
- 10 Level-2 EV Chargers & 1 Auto Car Wash Unit.

Out-of-Scope Components

- Major structural civil modifications to main slab.
- Municipal grid station capacity expansion.
- Customer vehicle pre-existing damage insurance.
- Long-term routine facility security payroll.

2. BUDGET ASSUMPTIONS & ESTIMATION METHODOLOGY

Estimates are formulated using a hybrid bottom-up and parametric method using Egyptian Pounds (EGP) as the primary base currency. Costs reflect mid-market industrial automation components adjusted for 15% import tariffs and local freight. All figures represent preliminary engineering estimates subject to supplier bidding.

3. COST BREAKDOWN STRUCTURE (CBS)

WBS	ITEM DESCRIPTION & SCOPE	QTY	UNIT	UNIT COST (EGP)	TOTAL COST (EGP)
1.0	**Autonomous Robots & Charging Docks** (Chassis, LiDAR, LiFePO4)	4	Set	1,200,000	4,800,000
2.0	**Sensors & IoT Infrastructure** (500 Bay Nodes, Gateways, Cabling)	1	Lot	1,150,000	1,150,000
3.0	**Mobile Application** (iOS/Android, UI/UX, Payment Integration)	1	Lot	720,000	720,000
4.0	**Central Control & Management System** (Fleet Core, Servers, Displays)	1	Lot	1,050,000	1,050,000
5.0	**AI Analytics & Prediction Engine** (Traffic Modeling, Dynamic Pricing)	1	Lot	600,000	600,000
6.0	**Drop-Off / Pick-Up Station Hubs** (Kiosks, Barriers, Scales, Light Curtains)	6	Hub	275,000	1,650,000

WBS	ITEM DESCRIPTION & SCOPE	QTY	UNIT	UNIT COST (EGP)	TOTAL COST (EGP)
7.0	**Value-Added Services** (10 Level-2 EV Chargers + 1 Car Wash System)	1	Lot	1,850,000	1,850,000
8.0	**Civil, Electrical & Safety** (Epoxy Floor, Electrical Panels, Fencing)	1	Lot	1,400,000	1,400,000
9.0	**Installation, Integration & Testing** (System Commissioning & Audits)	1	Lot	950,000	950,000
10.0	**Staff Training & Handover** (Documentation & Technical SOPs)	1	Lot	330,000	330,000
BASE CAPITAL EXPENDITURE (CAPEX) TOTAL					**15,000,000**

4. CAPITAL ALLOCATION & OPERATIONAL COSTS (OPEX)

CAPEX Distribution

- **Robotic Valets (AGVs):** 32.00% (4.80M EGP)
- **Smart Services (EV/Wash):** 12.33% (1.85M EGP)
- **Drop-Off Stations:** 11.00% (1.65M EGP)
- **Civil & Safety Works:** 9.33% (1.40M EGP)
- **Sensors & IoT Mesh:** 7.67% (1.15M EGP)
- **Software & AI Core:** 15.80% (2.37M EGP)
- **Integration & Training:** 8.54% (1.28M EGP)

Annual OPEX Summary (Year 1 Baseline)

OPERATING COST COMPONENT	ANNUAL (EGP)
Facility Electricity Draw	540,000
EV Charger & Wash Utilities	420,000
Cloud Server & API Licenses	180,000
Preventive AGV & Station Maint.	300,000
Battery/Sensor Replacement Fund	360,000
On-Site Tech Support Personnel	720,000
Software Maintenance & Updates	144,000
TOTAL ANNUAL OPEX	**2,664,000**

5. RISK CONTINGENCY & MANAGEMENT RESERVES

Reserves prevent financial exhaustion caused by design changes, import tariffs, or integration delays.

```
Base CAPEX = 15,000,000 EGP
Contingency Reserve (15% for Known-Unknowns) = 0.15 × 15,000,000 = 2,250,000 EGP
Project Cost Baseline = 15,000,000 + 2,250,000 = 17,250,000 EGP
Management Reserve (10% for Unknown-Unknowns) = 0.10 × 15,000,000 = 1,500,000 EGP
Total Authorized Project Budget = 17,250,000 + 1,500,000 = 18,750,000 EGP
```

6. COST CONTROL & FINANCIAL FEASIBILITY ANALYSIS

Cost performance is tracked bi-weekly using Earned Value Management (EVM). Key metrics include:

- **Cost Variance:** `CV = EV - AC` (Target: CV ≥ 0)
- **Cost Performance Index:** `CPI = EV / AC` (Threshold Alert: CPI < 0.90 triggers compulsory engineering review)

5-Year Discounted Cash Flow (DCF) Metrics

Assuming 500 bays operating at 50% mean occupancy (15 EGP/hr) plus VIP, EV, and washing revenues, projected annual gross revenue is **26,097,500 EGP**. Factoring an 18% Discount Rate (WACC) and 5% annual OPEX inflation:

> • **5-Year Net Present Value (NPV):** 35,272,930 EGP (Highly Viable)
> • **Internal Rate of Return (IRR):** 124.5% | **Simple Payback Period:** 0.74 Years (~9 Months)

7. RECOMMENDATIONS & NEXT STEPS

1. Issue RFP competitive bidding for the 4 AGV units to replace initial supplier baseline quotes.
2. Conduct a sub-scale prototype field test (1 AGV, 10 bays) to freeze software specs before full release.